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WRF Cold-Pool Sensitivities Relevant to Winter-Ozone Modelling in the Uinta Basin

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Abstract

Cold-pool structure controls the confinement and lower-atmospheric flow relevant to wintertime ozone in Utah's Uinta Basin, but surface observations alone cannot resolve that structure. We evaluated Weather Research and Forecasting (WRF) simulations of 2 February 2013 for sensitivity to Global Forecast System (GFS) or North American Mesoscale (NAM) forcing, nest feedback, boundary-layer and surface-layer schemes, slope-aware radiation, and static terrain. Across five matched hours, GFS-driven runs were warmer at the evaluated surface sites, had an approximately 3.6 K smaller 50–500 m potential-temperature endpoint difference, and a 0.8–1.0 m s^{-1} larger layer-mean wind-vector magnitude than feedback-matched NAM runs. GFS also retained less footprint-mean heat deficit under all seven definitions. Every run gained heat deficit from 1400 to 1800 UTC, despite within-window declines, so the product separation represents a state difference without demonstrated faster cold-pool loss. Feedback and physics responses were smaller and metric dependent; terrain diagnostics revealed spatial responses that the station sample could miss. The results guide meteorological diagnostics for subsequent air-chemistry modelling, while leaving the initial-versus-lateral forcing contribution, inversion depth and basin export unresolved.

Keywords: winter ozone; WRF simulation; meteorological uncertainty; cold-air pool; complex terrain; model resolution

1. Introduction

Wintertime ozone in the Uinta Basin develops within snow-covered, stable lower-atmospheric layers that confine precursors near the surface [1–3]. The basin floor, benches and surrounding terrain support strong temperature contrasts and thermally driven slope flows [4,5]. Comparable winter ozone episodes occur in Wyoming's oil-and-gas-producing basins [6]. Longer Uinta records show changing ozone and precursor conditions [7]; here we examine the meteorology of one 2013 episode over five retained hours.

Surface observations alone cannot validate inversion depth or the three-dimensional mixing volume supplied to a chemistry model. Evaluation therefore requires complementary diagnostics of potential temperature, lower-layer winds and heat storage, with a clear distinction between observed agreement and modelled structure.

Earlier Uinta modelling demonstrates strong case dependence. Snow cover can strengthen cold pools by increasing albedo, while cloud phase alters longwave forcing and nocturnal cooling [2]. Snow, microphysics and grid choices can thus alter both the cold pool

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and its representation on evaluation and chemistry grids. Four-dimensional data assimilation (FDDA) depends on the nudged variables, input analysis and scheme-dependent boundary-layer diagnosis [8]. A subsequent agency study examined reinitialisation, coordinates, snow initialisation and nudging, but assimilated evaluation data and differing samples limit configuration comparisons [9]. These studies motivate the present sensitivity tests.

Meteorological, emissions, photolysis, deposition and chemical-mechanism errors can compensate in coupled simulations. Changing emissions with fixed meteorology can improve ozone agreement without uniquely identifying inventory error [1]. Cumulative changes to these processes complicate attribution, and improved temperature profiles need not improve ozone profiles [10]. Credible meteorology reduces this confounding but does not guarantee accurate ozone.

We test whether the GFS–NAM lower-atmospheric response recurs under one-way and two-way nest feedback, and compare it with physics, radiation and terrain responses. The diagnostics combine surface skill, the 50–500 m potential-temperature endpoint difference, layer-mean horizontal wind-vector magnitude and footprint-mean heat deficit. Together they describe the simulated cold-pool state; they do not measure inversion depth, ventilation or basin export.

We ask:

1. Does the GFS–NAM lower-layer response recur under both feedback modes?
2. How do feedback, boundary-layer and surface-layer schemes, and slope-aware radiation compare with the forcing-product response?
3. How does changing the static-terrain source affect the atmosphere on the same WRF grid?
4. What do the combined diagnostics reveal that a single station metric can miss?

The study contains no chemistry calculation and does not separate initial from lateral-boundary forcing. Planned follow-on work led by L.D. will connect meteorological representation with air-chemistry modelling for this and related cases.

2. Materials and Methods

2.1. Case, Observations, and Evaluation Domains

The case falls within the January–February 2013 period of snow cover, cold pools and elevated Uinta ozone [1,2,8]. WRF integrations cover 1200–1800 Coordinated Universal Time (UTC), or 0500–1100 Mountain Standard Time (MST; UTC−7), on 2 February. Excluding the initialised output and first-hour forecast leaves the longest common window, 1400–1800 UTC (0700–1100 MST), with five hourly samples. Synoptic Data PBC Mesonet observations provide 2 m temperature at 11 sites, 10 m wind speed at 10 sites and surface pressure at 9 sites [11]. The temperature sample includes every registered station reporting in the window; Sand Wash had no reports. Subhourly reports were averaged within each UTC clock hour. The stations span the basin floor, sidewalls and benches but are unevenly distributed and provide no independent free-air vertical validation. We used the 1 km d02 domain because seven stations lie outside the usable d03 interior (Figure 1); a common-support d02–d03 comparison remains future work. Four stations provide context from 31 January 0000 UTC to 4 February 0000 UTC. For the observed and modelled Mountain Home-minus-Pariette Draw index, we calculated surface potential temperature as $\theta = T(1000/p)^{0.2854}$, with T in kelvin and p in hPa, using the dry-air Poisson relation [12]. This horizontally separated bench-minus-floor index measures a surface contrast, not free-air inversion strength or depth.

Ozone context uses the US Environmental Protection Agency (EPA) Air Quality System (AQS) AirData daily archive for January–February 2013 [13]. We retained the "Ozone 8-hour 2015" daily-summary row at Ouray, Redwash, Roosevelt, Vernal and Little Mountain: one row per site on each of 59 local dates, with within-day coverage of 47–100% marked in the figure. The satellite panel uses the National Aeronautics and Space Administration (NASA) Global Imagery Browse Services (GIBS) Terra Moderate Resolution Imaging Spectroradiometer (MODIS) corrected-reflectance 7-2-1 layer for 2 February [14]. Interpretation uses the spectral absorption and scattering principles described by Liou [15], with snow/cloud colours specified by NASA's layer documentation [16].

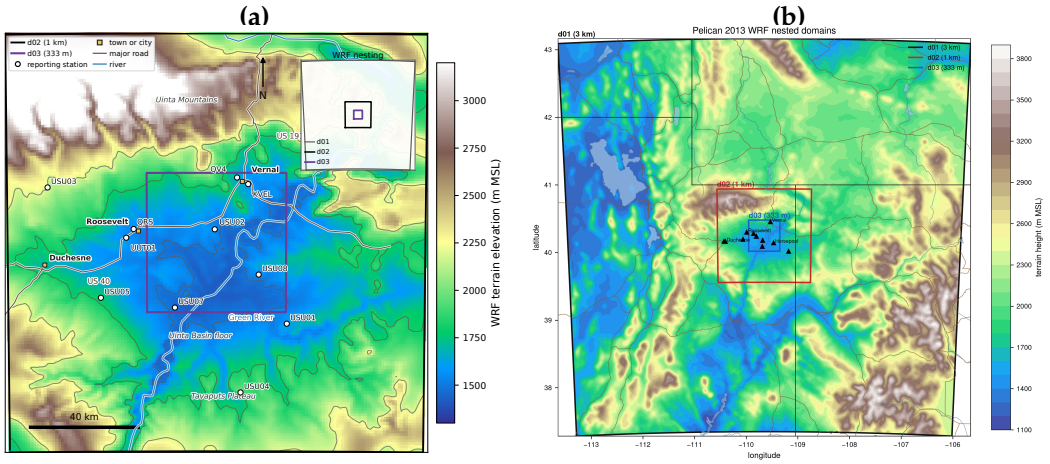


Figure 1. Geographic and sampling orientation for the evaluation. (a) The d02 terrain, reporting stations, and d03 interior define the common evaluation setting. (b) The d01–d02–d03 WRF grid layout places that setting within the full three-domain configuration. The d02 domain contains all eleven reporting stations; d03 contains four usable station columns.

2.2. WRF Configuration and Archived Simulation Family

We used WRF version 4.8.0 [17], initialised at 1200 UTC with lateral boundary conditions through 1800 UTC on 2 February 2013. The inputs were the 1200 and 1800 UTC zero-hour GFS grid 004 (0.5°; `gfsanl_4`) or NAM grid 218 (12 km; `namanl_218`) analyses [18,19]. These retrospective integrations include temporally interpolated lateral-boundary forcing, so the retained response combines initial-state and boundary influences. Three nested domains, d01–d03, have horizontal spacings of 3 km, 1 km and 333.333 m, with 75 eta levels, approximately 20 m spacing near the surface and a 50 hPa model top. The reference physics comprises Thompson microphysics [20], Rapid Radiative Transfer Model for General Circulation Models (RRTMG) longwave and shortwave radiation [21], the revised fifth-generation Pennsylvania State University–National Center for Atmospheric Research Mesoscale Model (MM5) similarity surface layer [22], the Noah land-surface model [23] and the Yonsei University (YSU) planetary boundary layer (PBL) scheme [24]. Grid and observation FDDA were disabled. Table 1 lists the simulations analysed here.

Static terrain entered through the WRF Preprocessing System and was fixed within each non-terrain comparison. The fine, default and coarse sources have spacings of 3 arc seconds, 30-arc-second class and 5 arc minutes, respectively; the coarse source is GMTED2010 [25]. These correspond to approximately 90 m, 900 m and 9 km north–south spacing at 40°N, distinct from WRF grid spacing. Snow and other land-surface initial fields entered through product-specific preprocessing, so their contributions remain part of the GFS–NAM treatment.

Table 1. Simulations analysed in this study. Every run uses WRF 4.8.0, three nests at 3/1/0.333 km with 75 levels, Thompson microphysics, RRTMG radiation, and Noah land surface unless stated. "Fine terrain" is the custom 3-arc-second source, "default terrain" is 30-arc-second class, and "coarse terrain" is the 5-arc-minute GMTED2010 source; static-source spacing is not model-grid spacing.

Code	Product, feedback	Terrain and physics
X0	NAM, one-way	fine terrain, YSU/MM5
X1	NAM, one-way	X0 plus slope radiation and shading
X2	NAM, one-way	X1 with MYJ/Eta instead of YSU/MM5
X3	NAM, one-way	default-terrain reference configuration
X4	NAM, one-way	deliberately coarse 5-arc-minute terrain
X5	NAM, two-way	default terrain
X6	GFS, two-way	default terrain
X9	NAM, two-way	X0 with two-way feedback
X10	GFS, one-way	default terrain

2.3. Experiment Comparisons

The primary comparison crosses GFS and NAM forcing with one-way and two-way feedback on default terrain (X3, X5, X6 and X10). GFS and NAM inputs supply 27 and 40 metgrid levels, respectively. Additional pairs test boundary-layer and surface-layer schemes, slope radiation, feedback on fine terrain, and terrain source; each is introduced with its result. Within each pair, the evaluation domain, station or footprint sample, valid hours and diagnostic calculation are matched.

2.4. Surface and Common-AGL Diagnostics

Surface variables and vertical fields were sampled from the same nearest d02 column at each station, keeping the model terrain and profile consistent. Station-to-cell distances are 133–594 m, with default-terrain elevation offsets of −32.4 to +14.3 m (Appendix Table A6). These pairings retain differences in exposure, slope and land cover; alternative interpolation methods were not tested. Mass-level height above ground level (AGL) was calculated from perturbation plus base geopotential relative to local terrain. Winds were destaggered and rotated to earth-relative components. Potential temperature was interpolated at the layer endpoints; wind components were linearly interpolated every 25 m from 50 to 500 m AGL for height integration. We repeated the four-run, eleven-column, five-hour comparison for 50–250 and 100–500 m layers and 10 and 50 m wind integration increments. The bounds define a common shallow layer rather than an observed inversion boundary. The increments test sampling of the same interpolated winds; they leave the potential-temperature endpoints unchanged and do not test native vertical resolution or interpolation method.

The finite-layer potential-temperature endpoint difference was defined as

$$S_{50:500} = \theta(500\,\mathrm{m}) - \theta(50\,\mathrm{m}) \tag{1}$$

An increase of potential temperature with height indicates dry static stability [12,26], but the endpoint difference cannot resolve inversion height, sharpness or internal layering.

The magnitude of the layer-mean horizontal wind vector was defined as

$$U_{50:500} = \left(\bar{u}^2 + \bar{v}^2\right)^{1/2}, \tag{2}$$

where the bars denote height means from 50 to 500 m AGL, not temporal or turbulent averages. We also calculated the layer-mean scalar speed $\overline{(u^2+v^2)^{1/2}}$ and the vector-to-scalar ratio to identify directional cancellation. These state diagnostics do not measure

basin-boundary flux or pollutant residence time. Native WRF PBL height serves as a scheme-dependent companion diagnostic [8].

2.5. Pairing, Aggregation, and Short-Sample Sensitivity

Run differences were paired by station and valid hour, then reduced to one unweighted spatial median per hour. Reported five-hour effects are means of those medians; stations are not independent replicates. To explore temporal dependence [27], a circular moving-block calculation drew 20,000 resamples of the five medians using two-hour blocks. Its 2.5th and 97.5th percentiles are descriptive sensitivity bounds, not confidence intervals for this single deterministic case.

2.6. Footprint-Integrated Heat Deficit

We adapted the basin heat-deficit formulation of Whiteman et al. [28] and the column implementation of Adler et al. [29]. Using hydrostatic balance, $dp = -\rho g\, dz$, to express column mass in pressure coordinates [12], we defined

$$H = \frac{c_p}{g} \int_{p_c}^{p_s} \max\{\theta_{\text{crest}} - \theta(p), 0\}\, dp, \tag{3}$$

where c_p is dry-air specific heat at constant pressure, g is gravitational acceleration, and $p_s > p_c$ are surface and crest-reference pressures. The reference potential temperature is interpolated to the fixed crest height in each modelled column; clipping retains only positive deficits. This potential-temperature proxy describes stratification relative to that reference, rather than an exact energy requirement for removing the cold pool. We report its cell-area-weighted footprint mean in MJ m^{-2}. The primary footprint is the convex hull of eleven stations and eleven basin waypoints, restricted to terrain below 2200 m: 4,965 d02 cells covering 5,277 km^2. Seven predeclared definitions use this hull at 2100, 2200 and 2300 m; the 2200 m hull dilated or eroded by one cell; and station-only and waypoint-only hulls at 2200 m. Feedback-matched GFS–NAM pairs share a terrain mask. The weaker-state criterion requires a negative GFS-minus-NAM difference at every retained hour under all seven definitions. The faster-loss criterion requires GFS heat deficit to decline from 1400 to 1800 UTC and to decline more than NAM under all seven definitions. Both criteria preceded output inspection; 1300 UTC is trajectory context only. The modelled footprint is an evaluation region, not a measured or hydrologically defined basin volume.

3. Results

3.1. Observed Episode Context

The MODIS image shows extensive snow and ice across the basin and surrounding terrain (Figure 2a). The AQS series places 2 February within a broader period of elevated ozone, with daily maximum eight-hour concentrations of 56–95 ppb across the five sites (Figure 2b). These observations establish episode context without attributing ozone to the simulated meteorology.

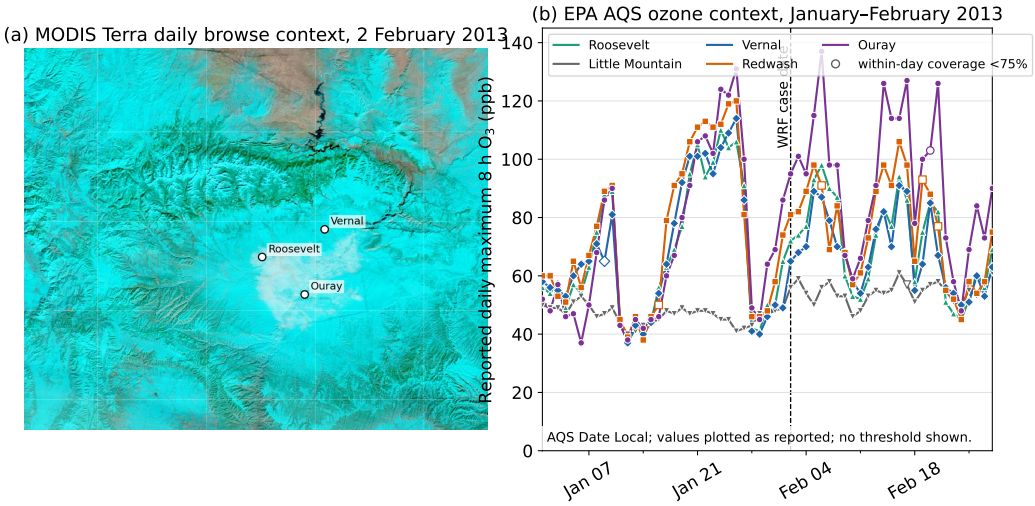


Figure 2. Observed context for the 2 February 2013 simulation date. (a) NASA GIBS Terra MODIS corrected reflectance, bands 7-2-1, for the 2 February calendar day; snow and ice generally appear cyan, most liquid cloud appears white, and ice cloud may also appear cyan. The daily browse layer provides qualitative snow/cloud context; the nearest centre-covering Terra granule was acquired at 1815–1820 UTC. (b) EPA AQS first maximum value from the "Ozone 8-hour 2015" daily-summary row at Ouray, Redwash, Roosevelt, Vernal, and Little Mountain, converted from ppm to ppb and plotted by AQS local date without recomputation. All five sites have 59 daily rows; hollow markers identify the seven site-days with less than 75% within-day observation coverage. The panel provides episode context, not a chemistry evaluation.

3.2. Observed Surface-Temperature Context

The basin-floor station repeatedly separated from the higher sites between 31 January and 3 February, with diurnally varying temperature differences (Figure 3). Seep Ridge (USU04; 1933 m) has a distinct trace despite a station-to-cell distance of only 133 m and a +2.9 m terrain offset. Exposure, slope flow and representativeness are possible explanations, but the metadata do not distinguish them. A fresh retrieval returned all 17 quarter-hour samples for temperature, humidity, wind speed and direction without provider QC flags, but no sensor-position or exposure history. It neither independently validates the measurements nor reproduces historical-cache processing. The recurring floor-to-bench separation is consistent with a cold-pool surface pattern; the horizontally separated sites cannot determine its vertical structure.

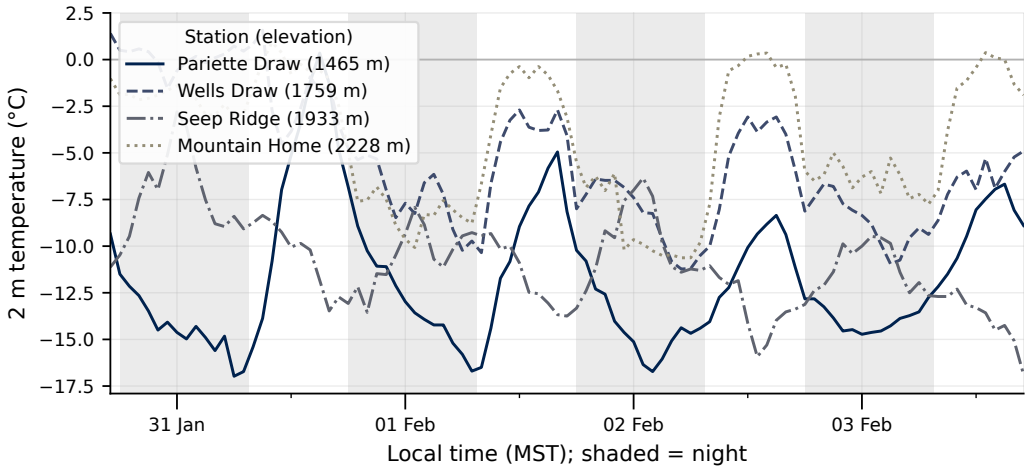


Figure 3. Observed 2 m-temperature evolution at four elevation-spanning Uinta Basin stations from 31 January through 3 February 2013. Shading indicates approximate local night. The series show a spatial surface-temperature pattern, not a colocated vertical profile.

3.3. Overview of Experiment Responses

The forcing-product contrasts give the largest temperature, endpoint-difference and layer-mean wind responses (Figure 4). The PBL/surface-layer response is smaller; slope-radiation, feedback and product-by-feedback interaction effects cluster near zero on these scales. Pressure and wind-error metrics have different orderings, preventing an overall configuration ranking.

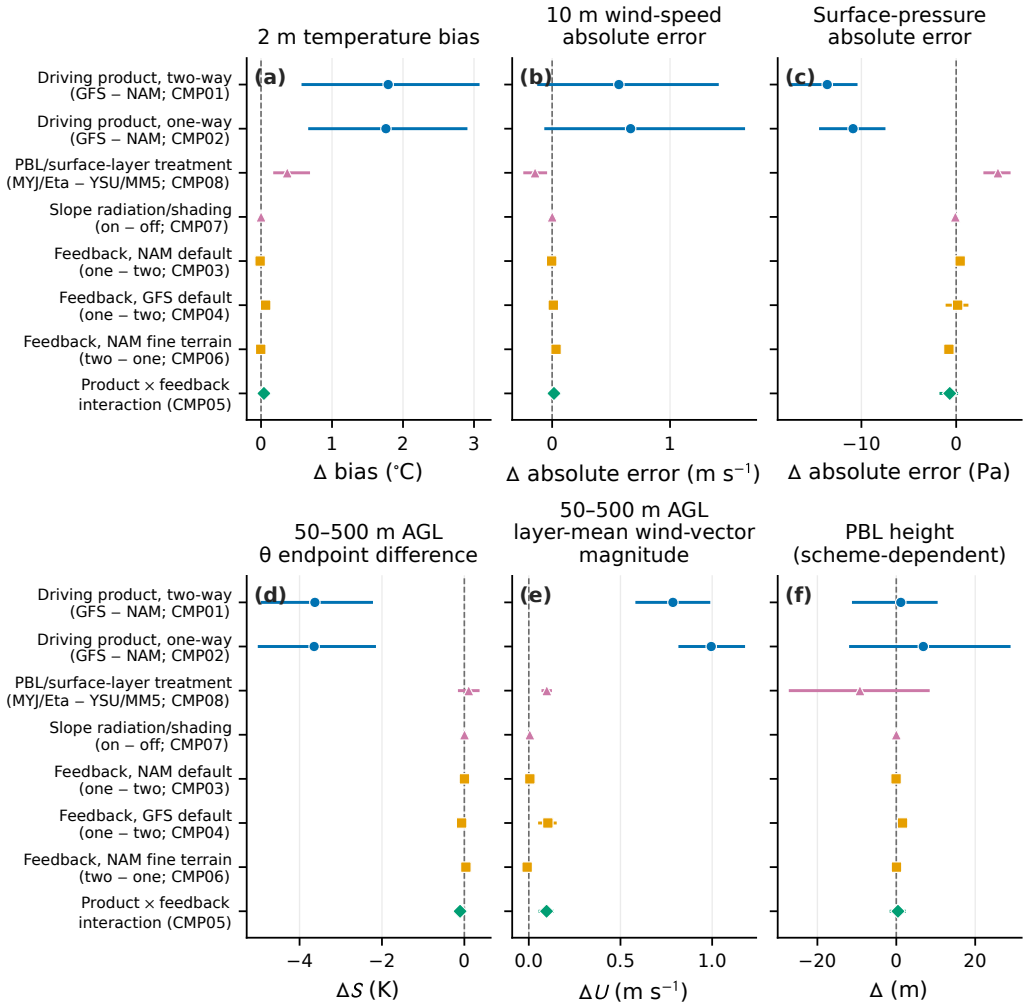


Figure 4. Mean five-hour effects for the controlled comparisons that share the station/common-AGL pipeline. Colours and marker shapes identify forcing-product, process-treatment, feedback, and interaction contrasts; each row states its differencing direction. Whiskers show descriptive block-sensitivity intervals for five dependent hourly medians, not confidence intervals.

3.4. Driving-Product Response in the Common Lower Layer

GFS produced a smaller 50–500 m potential-temperature endpoint difference than NAM under both feedback modes (Figure 5). Paired GFS-minus-NAM effects averaged −3.63 K with two-way feedback and −3.65 K with one-way feedback, retaining their sign at every hour (Table A1). Layer-mean wind-vector magnitude increased by 0.79 and 0.99 m s^{-1}, respectively, also with consistent hourly signs. Scalar-speed differences were similar, so directional cancellation does not create the speed ordering (Appendix Table A2). The vector-to-scalar ratio difference changed sign in two of five hours for each pair.

Temperature and wind retain their hourly orderings across all nine diagnostic configurations. Their magnitudes depend on layer bounds (Appendix Table A3), whereas changing the integration increment at fixed bounds alters the mean wind effects by less than 0.02 m s^{-1}. Native PBL-height differences change sign under both feedback modes, consistent with the diagnostic's scheme dependence [8].

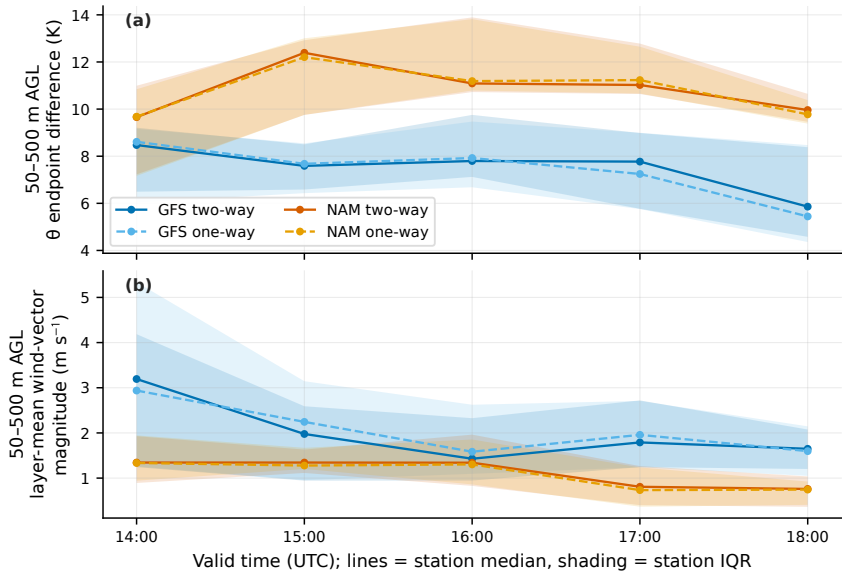


Figure 5. Station-column medians and interquartile ranges for the 50–500 m AGL potential-temperature endpoint difference and layer-mean wind-vector magnitude over the matched window. Solid lines denote two-way feedback and dashed lines denote one-way feedback. Both feedback modes retain the GFS–NAM ordering throughout the window.

3.5. Surface and Spatial Corroboration

GFS was warmer at the surface than NAM, consistent with the common-layer diagnostics (Table A4). The 2 m temperature-bias contrasts were approximately +1.8 °C under both feedback modes. Pressure absolute error decreased, while hourly wind-error differences changed sign. Figure 6 compares the observed two-site surface index with the two-way simulations.

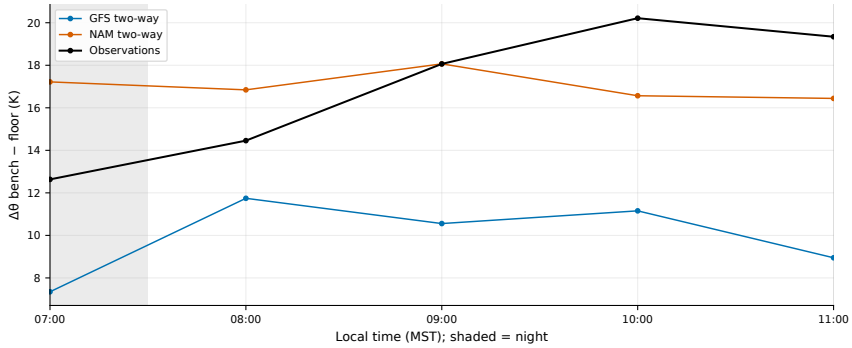


Figure 6. Observed and modelled Mountain Home-minus-Pariette Draw surface potential-temperature contrast from 1400 to 1800 UTC on 2 February 2013. The modelled series are X6 (GFS two-way) and X5 (NAM two-way) on d02, sampled at the same five hours as the observations. The index is a horizontally separated bench-minus-floor surface contrast.

The five-hour mean Mountain Home-minus-Pariette Draw contrast was 16.94 K observed, 17.03 K in NAM and 9.95 K in GFS. NAM's agreement supports this surface index without validating inversion depth or overall skill. Across the footprint, GFS-minus-NAM near-surface responses were positive on average and concentrated below 500 m AGL under both feedback modes (Table A4). The spatial-coherence test nevertheless failed: only 64–67% of cells had the expected sign at 1400–1500 UTC, below the required two-thirds share at every hour. Later coherence does not change that outcome.

3.6. Footprint Heat Deficit

GFS retained less footprint-mean heat deficit at every hour under all seven definitions, with mean paired effects of approximately $-1.2\,\mathrm{MJ\,m^{-2}}$ under both feedback modes (Table A5). Every run ended with more deficit at 1800 than at 1400 UTC, despite within-window declines (Figure 7). The weaker-state criterion therefore passed and the faster-loss criterion failed.

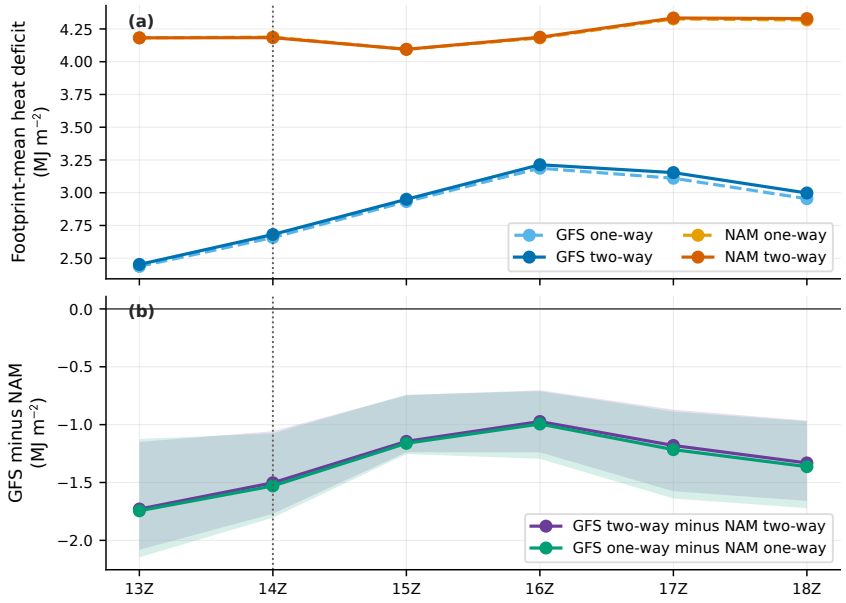


Figure 7. Footprint-mean heat-deficit trajectories for the four default-terrain matrix members and seven-definition sensitivity envelopes for the two GFS-minus-NAM pairs. The 1300 UTC values orient the trajectories but are excluded from every reported criterion and effect. GFS retains a smaller deficit throughout, but the deficit does not decline faster in either GFS member.

3.7. Process Treatments on the Fine-Terrain Branch

Replacing YSU/revised-MM5 with Mellor–Yamada–Janjić (MYJ)/Eta schemes [30] on fine terrain with slope radiation (X2 minus X1) increased temperature bias by 0.37 °C, reduced wind absolute error by $0.145\,\mathrm{m\,s^{-1}}$ and increased layer-mean wind-vector magnitude by $0.098\,\mathrm{m\,s^{-1}}$. All changes retained their sign across the five hours but were smaller than the forcing-product effects on shared lower-layer metrics. The opposing temperature- and wind-error changes preclude an overall scheme ranking.

Adding slope radiation and shading (X1 minus X0) had negligible effects at basin-floor station columns but produced a spatially organised sidewall skin-temperature response. The archived flat-plane shortwave field and missing surface heat fluxes prevented a direct mechanism test or configuration recommendation.

3.8. Nest-Feedback Robustness across Two Terrain Regimes

Default-terrain feedback effects were small for both products: one-way minus two-way contrasts were at most 0.07 K in endpoint difference and $0.10\,\mathrm{m\,s^{-1}}$ in wind-vector magnitude (Table A1). The fine-terrain NAM comparison (X9 minus X0, two-way minus one-way) was similarly small, with block-sensitivity bounds spanning zero for both metrics. Surface wind and pressure changes retained their signs but were also small, providing no feedback recommendation beyond this case.

3.9. Terrain-Source Brackets

The terrain comparisons hold NAM forcing, one-way feedback, WRF grid spacing and YSU/MM5 physics fixed. Default minus fine (X3 minus X0) had a small d02 response and failed its predeclared terrain-response test. The wider default-minus-coarse (X3 minus X4) and fine-minus-coarse (X0 minus X4) brackets gave larger spatial responses. For fine minus coarse, the heat-deficit median changed sign between d02 (+0.014 MJ m^{-2}) and d03 (−0.184 MJ m^{-2}), while the two d03 section medians were +0.278 and +0.369 K. Figure 8 shows this domain-dependent response without establishing terrain-source superiority, drainage pathways or basin cold-air-volume change.

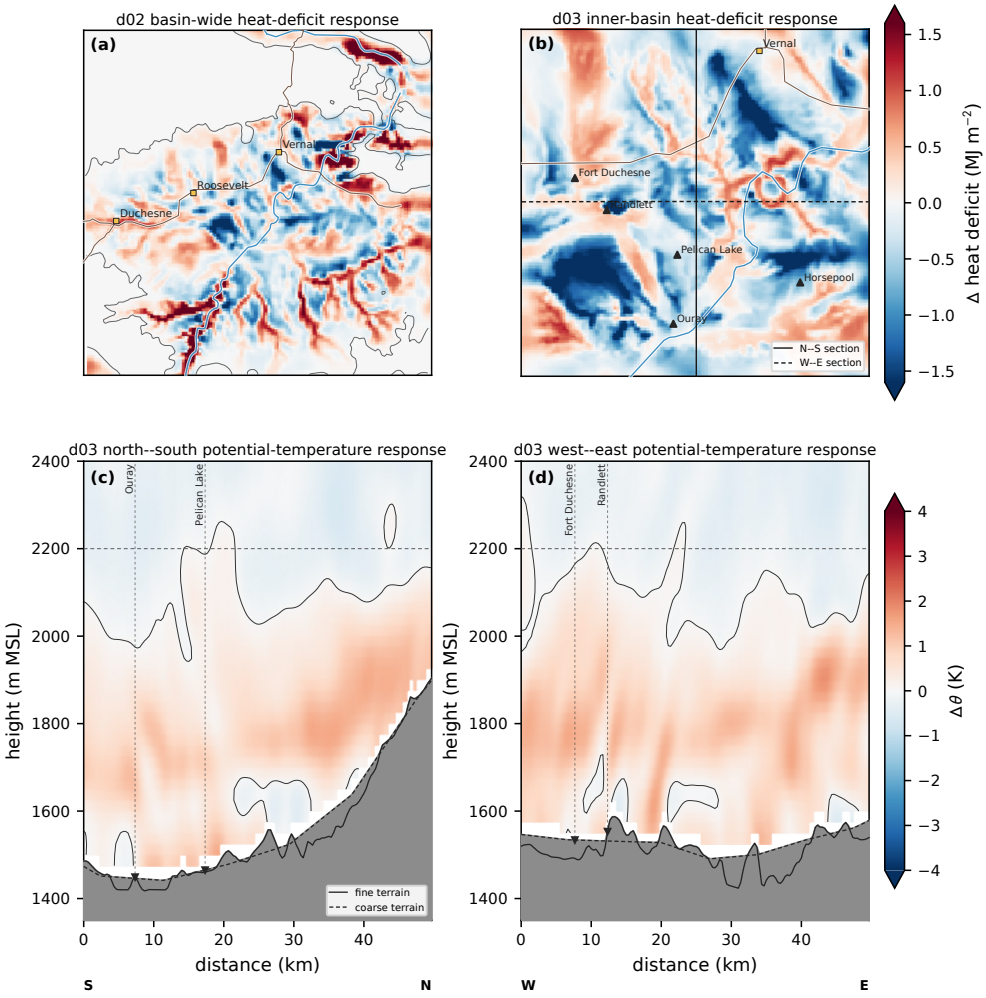


Figure 8. Diagnostic response to the widest controlled static-terrain bracket at 1600 UTC, calculated as X0 minus X4 (fine 3-arc-second minus coarse 5-arc-minute source). (a) The basin-wide d02 and (b) inner-basin d03 top-down heat-deficit maps share a fixed ±1.6 MJ m^{-2} scale. (c) The north–south and (d) west–east d03 lower-atmosphere potential-temperature sections share a fixed ±4 K scale and the 2200 m crest reference. Panels retain their domain-specific spatial support.

4. Discussion

4.1. Forcing-Product Differences in the Lower Layer

The principal result is the repeated GFS–NAM separation under both feedback modes: GFS has a smaller potential-temperature endpoint difference, larger layer-mean wind-vector magnitude and less heat deficit. These diagnostics are consistent with weaker confinement, while pressure and wind errors prevent an overall skill ranking. Because confinement, emissions, snow, cloud, photolysis and deposition errors can compensate

in coupled simulations [1,2,10], forcing-product sensitivity deserves attention in a future chemistry ensemble. Its importance across other episodes remains to be tested.

The modelled d03 Horsepool profiles show product-specific initial states and first-hour evolution (Figure 9). They combine early model adjustment with lateral-boundary evolution and lack colocated observed validation.

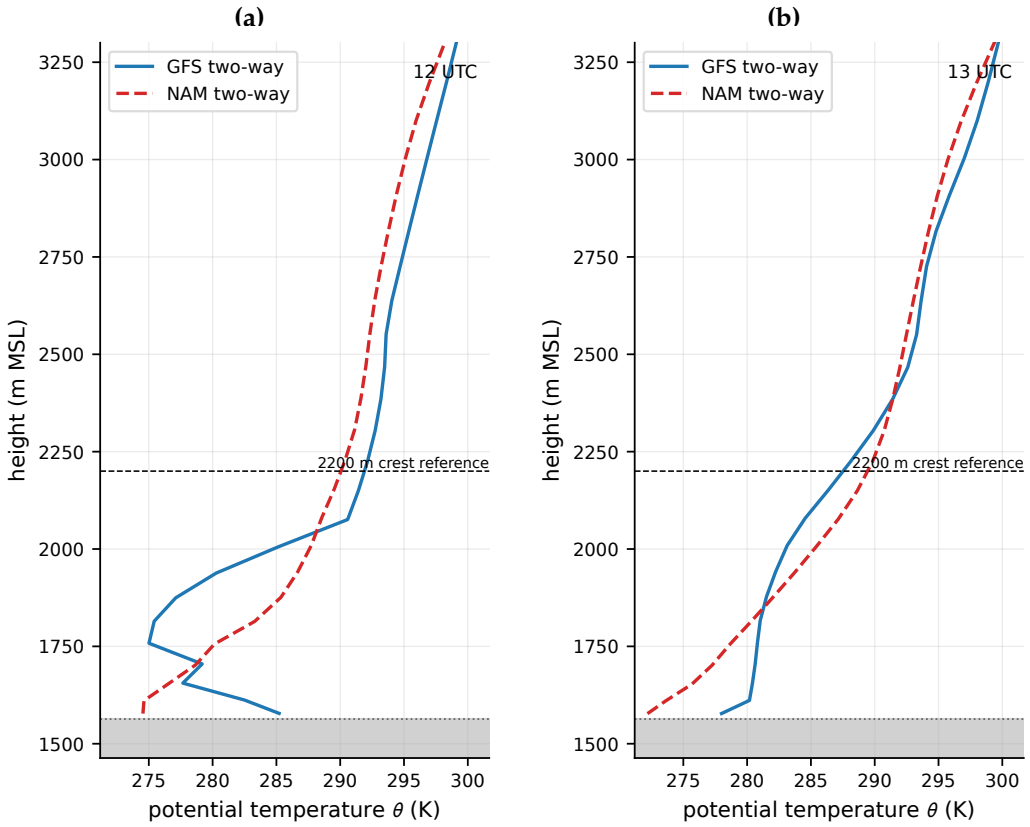


Figure 9. Model-only d03 potential-temperature profiles at Horsepool for X6 (GFS two-way) and X5 (NAM two-way). (a) The 1200 UTC initialised states and (b) the 1300 UTC one-hour forecasts use identical axes. Grey shading marks levels below the local model terrain, and the dashed 2200 m line is the fixed heat-deficit crest reference rather than an observed cold-pool top. Changes between panels include model adjustment and lateral-boundary evolution.

4.2. Heat-Deficit Evolution and Spatial Extent

Lower GFS heat deficit persisted across all definitions, but positive net changes in every run rule out demonstrated faster loss during the retained window. At least a full diurnal cycle is needed to test differences in cold-pool erosion or persistence. The forcing-product response was strongest below 500 m AGL and consistent in its footprint mean, while the failed every-hour spatial-coherence test limits its spatial generality.

4.3. Physics, Feedback and Terrain Sensitivities

Physics and feedback effects were smaller than the forcing-product response on the common temperature and wind diagnostics. The scheme treatment improved wind error while increasing temperature bias; feedback remained small on both terrain branches. Slope radiation and terrain changes produced spatial responses that station metrics could miss. These results favour combining station and spatial diagnostics, while leaving radiation and terrain-source recommendations unresolved.

4.4. Observations and Future Experiments

Independent vertical profiles are needed to evaluate inversion depth, layer stability and heat deficit; neither surface contrasts nor scheme-dependent native PBL height provide that validation. A bounded search of accessible holdings found no measured in-basin profile within the retained window. The University of Utah's provisional Roosevelt sounding began at 1801:54 UTC, just after the window, and was neither quality controlled nor compared with WRF here [31]. Using it requires an explicit time tolerance, quality control and matched analysis. Further work should separate initial from lateral-boundary forcing, test common-support d02–d03 responses, compare nudging with independent verification, and evaluate multiple cold-pool episodes.

5. Conclusions

The GFS–NAM contrast produced the largest repeated lower-layer response on the common d02 sample. Under both feedback modes, GFS was warmer at the evaluated surface sites, had an approximately 3.6 K smaller 50–500 m potential-temperature endpoint difference, a 0.8–1.0 m s^{-1} larger wind-vector magnitude and approximately 1.2 MJ m^{-2} less footprint-mean heat deficit. Scalar speed retained the wind ordering, while varying directional alignment supports no transport interpretation. The heat-deficit separation represents a state offset without demonstrated faster loss; surface errors do not yield an overall product ranking.

Feedback effects were small (at most 0.07 K and 0.10 m s^{-1} on the same layer metrics), while MYJ/Eta increased temperature bias by 0.37 °C and reduced wind absolute error by 0.145 m s^{-1}. Slope radiation and terrain effects depended on spatial support: the fine-minus-coarse heat-deficit median changed from +0.014 MJ m^{-2} on d02 to −0.184 MJ m^{-2} on d03. These findings support complementary station, layer and footprint diagnostics for meteorology supplied to chemistry models. Longer matched integrations and independent vertical observations are needed to identify forcing contributions and evaluate inversion depth and persistence; this study establishes neither basin export nor ozone response. Appendix A retains the numerical summaries and sampling metadata.

Appendix A. Data Tables and Secondary Diagnostics

Table A1. Paired effects for the default-terrain forcing-product/feedback matrix, averaged over the five hourly spatial medians. Positive temperature values indicate a warmer model-minus-observation bias, positive endpoint-difference values indicate a larger 50–500 m potential-temperature difference, and positive wind-vector values indicate a larger layer-mean wind-vector magnitude. Values are descriptive means for this case.

Effect	ΔT2 bias (°C)	$\Delta S_{50:500}$ (K)	$\Delta U_{50:500}$ (m s^{-1})
GFS − NAM, two-way	+1.79	-3.63	+0.79
GFS − NAM, one-way	+1.76	-3.65	+0.99
GFS one-way − two-way	+0.07	-0.07	+0.10
NAM one-way − two-way	-0.01	+0.00	+0.01
Feedback interaction	+0.04	-0.10	+0.10

Table A2. Primary-layer wind-metric comparison for the feedback-matched GFS-minus-NAM pairs. Each value is the mean of five hourly paired-station medians. The alignment ratio is the magnitude of the layer-mean vector divided by layer-mean scalar speed; its difference changed sign in two of five hours for each pair.

Feedback	Δ vector magnitude (m s^{-1})	Δ scalar speed (m s^{-1})	Δ alignment ratio
Two-way	+0.79	+0.78	+0.063
One-way	+0.99	+1.01	+0.053

Table A3. Layer-bound sensitivity of the GFS-minus-NAM contrasts at the primary 25 m wind integration increment. Each value is the mean of five hourly paired-station medians on d02 over the same eleven columns. ΔS is the potential-temperature endpoint-difference contrast; ΔU and ΔV are the layer-mean vector-magnitude and scalar-speed contrasts, respectively. All three layer choices also retain the hourly signs at 10 and 50 m integration increments.

	ΔS (K)		ΔU (m s^{-1})		ΔV (m s^{-1})	
Layer (m AGL)	Two-way	One-way	Two-way	One-way	Two-way	One-way
50–500	-3.63	-3.65	+0.79	+0.99	+0.78	+1.01
50–250	-2.22	-2.08	+0.67	+0.85	+0.70	+0.83
100–500	-3.38	-3.42	+0.77	+1.02	+0.79	+1.02

Table A4. Surface, spatial, and vertical corroboration for the two GFS-minus-NAM forcing-product contrasts. All values use the 1400–1800 UTC window described in Section 3.5. The first four rows use station samples; the last four use area-weighted diagnostics over the registered basin-floor footprint.

Diagnostic	Result	Scope
2 m temperature	GFS-minus-NAM bias contrast +1.79 °C two-way and +1.76 °C one-way; absolute-error differences +1.41 and +1.50 °C	Same ordering as the common-AGL endpoint difference
10 m wind	Bias contrast +0.91 and +0.96 m s^{-1}; hourly absolute-error differences change sign	No stable wind-skill ordering
Surface pressure	Absolute-error contrast -13.6 Pa two-way and -10.9 Pa one-way; negative in every hour	Opposite ordering from temperature, with small absolute differences
Two-site surface index	Observed 12.63–20.21 K; NAM 16.44–18.07 K; GFS 7.35–11.74 K under two-way feedback	Horizontally separated surface contrast, not inversion strength or depth
Mean 2 m potential-temperature difference	+0.89 to +1.16 K at 1400–1500 UTC; +2.00 to +2.74 K at 1600–1800 UTC	Positive at all ten pair-hours
Footprint sign share	0.64–0.67 at 1400–1500 UTC; 0.84–0.93 later	Early response is not spatially uniform
Low-to-aloft absolute-response ratio	1.45–3.12	Response concentrated in the lower atmosphere
50–500 m endpoint-difference contrast	-4.33 to -3.04 K	Negative at all ten pair-hours

Table A5. Footprint-mean, vertically integrated heat-deficit contrasts over the 1400–1800 UTC decision window. The primary definition uses the combined station-and-waypoint hull with a 2200 m crest (4965 cells, 5277 km^2); envelopes span all seven footprint definitions. The values are deterministic descriptive contrasts in MJ m^{-2}.

GFS − NAM quantity	Two-way pair	One-way pair
Mean difference (primary)	-1.23	-1.25
Hourly range (primary)	-1.50 to -0.97	-1.53 to -0.99
Hourly envelope (seven definitions)	-1.77 to -0.70	-1.80 to -0.71
GFS change, 1400–1800 UTC	+0.32	+0.30
NAM change, 1400–1800 UTC	+0.15	+0.13
Difference in change (primary)	+0.17	+0.17

Table A6. Case-window temperature stations and nearest d02 sampling audit. Coordinates and reported elevations are from the registered Synoptic Data PBC metadata; separation is great-circle distance to the sampled cell centre, and the final column is X3 default-terrain model height minus reported station elevation.

ID	Site	Latitude	Longitude	Elevation (m)	Cell distance (m)	Elevation offset (m)
KVEL	Vernal Airport	40.44295	-109.51273	1604	405	0.0
QRS	Roosevelt	40.29430	-110.00900	1588	354	-32.4
QV4	Vernal	40.46472	-109.56083	1666	408	-2.0
USU01	Seven Sisters	39.98130	-109.34540	1615	555	-3.0
USU02	Gusher	40.29350	-109.65750	1550	230	-5.0
USU03	Mountain Home	40.43190	-110.38210	2228	594	-5.8
USU04	Seep Ridge	39.75390	-109.54600	1933	133	+2.9
USU05	Wells Draw	40.06700	-110.15100	1759	187	+10.1
USU07	Pariette Draw	40.03460	-109.83010	1465	485	-0.1
USU08	USU Research Trailer	40.14370	-109.46718	1570	446	-4.2
UUT01	Uintah Basin Research Trailer	40.26560	-110.04120	1561	419	+14.3

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